Reflection-symmetric diagonal synthesis: resolving a conjecture of Huang, Kosugi, Nishi and Matsushita

Haobo Ma* Wenlin Zhang†

1 November 2026

Licensed under CC BY 4.0

Abstract

Combined with the upper bound of Huang, Kosugi, Nishi and Matsushita [J. Phys. Soc. Jpn. **93**, 054002 (2024); arXiv:2310.06676v2], this resolves the conjecture stated in their Sect. 6 that $2^{2n-1} + 2n - 2$ CNOTs are optimal in the worst case. For every $n \geq 1$, an explicit reflection-symmetric diagonal unitary on $2n$ qubits requires at least $2^{2n-1} + 2n - 2$ CNOT gates in any exact, ancilla-free circuit over $\{\mathrm{CNOT}, R_z\}$, even up to global phase. The hard instance has diagonal entry i at the all-zero and all-one strings and entry 1 elsewhere. Multiplicative character invariants force all nonzero even parities to occur on rotated wires. A birth-and-return argument then gives the additional linear term in the CNOT bound. We give a self-contained exposition of the proof in the Trureturing source at the cited revision.

1 The synthesis question

How many CNOT gates are needed in the worst case to synthesize a reflection-symmetric diagonal unitary? Huang et al. [1] construct circuits on $2n$ wires using $2^{2n-1}+2n-2$ CNOTs and conjecture (Sect. 6) that this count is worst-case optimal. The explicit hard instance below establishes the matching lower bound, by forcing even wire parities and counting the gates needed to restore the wires.

Put $m = 2n$ and index the computational basis by $x \in \mathbb{F}_2^m$. Write $\bar{x} = x + \mathbf{1}$. A real phase function is reflection-symmetric when $\theta(\bar{x}) = \theta(x)$ for every x. This is exactly the two-register symmetry in Eq. (1) of Huang et al. [1]: complementing each n-bit register index complements all m bits. Throughout,

$$R_z(\varphi) = \mathrm{diag}(e^{-\mathrm{i}\varphi/2}, e^{\mathrm{i}\varphi/2}), \qquad \varphi \in \mathbb{R}.$$

The allowed circuits are finite, act on these m wires only, and use distinct target and control wires for each CNOT. No ancillary qubits, measurements, other gates, or approximation are allowed.

Theorem 1. *For each $n \geq 1$, define $\theta_* : \mathbb{F}_2^{2n} \to \mathbb{R}$ by*

$$\theta_*(x) = \begin{cases} \pi/2, & x = 0 \text{ or } x = \mathbf{1}, \\ 0, & \text{otherwise}, \end{cases} \qquad D_* = \mathrm{diag}(e^{\mathrm{i}\theta_*(x)}).$$

*ChronoAI Pte Ltd, Singapore.

†National University of Singapore; ChronoAI Pte Ltd, Singapore.

Then $\theta_*(\bar{x}) = \theta_*(x)$. Every allowed circuit with matrix $U = cD_*$, for any complex scalar c, contains at least

$$N_m = 2^{m-1} + m - 2 = 2^{2n-1} + 2n - 2$$

CNOT gates.

The scalar is automatically nonzero, indeed of unit modulus, because U and D_* are unitary. Theorem 1 is existential with respect to the class of symmetric targets: it does not say that every member of that class is hard. The identity is an immediate easy member. The lower-bound statement and proof are already present in the Trureturing formalization [4].

Parity networks give a standard description of CNOT/phase circuits. Amy, Azimzadeh and Mosca [2], Sect. 2, define these networks with a specified final linear map. Identity parity networks return that map to the identity.

Cao et al. [3] study the graphic case, where the required labels have weight two. For a connected graph their lower bound is $|E| + |V| - 1$. It gives 9 for the complete graph on four wires. Here the weight-four label is also forced, and the lower bound is 10. The argument below treats every nonzero even label and controls the freedom to change real phases by multiples of 2π.

2 Wire labels and multiplicative characters

Represent a wire's current Boolean value as $w \cdot x$, with label $w \in \mathbb{F}_2^m$. Initially the labels are the standard basis vectors e_j. A CNOT with target t and control c replaces w_t by $w_t + w_c$. Thus the label matrix stays invertible and its rows never vanish. A rotation on a wire labelled w multiplies the basis amplitude by

$$g(w \cdot x), \qquad g(0) = e^{-\mathrm{i}\varphi/2}, \quad g(1) = e^{\mathrm{i}\varphi/2}.$$

Induction over the circuit consequently gives a monomial normal form

$$|x\rangle \longmapsto f(x)|Ax\rangle, \qquad f(x) = \prod_k g_k(w_k \cdot x), \qquad A \in \mathrm{GL}(m, \mathbb{F}_2).$$

All factors are nonzero. If the circuit is diagonal up to a scalar, its nonzero entry on each input cannot change the basis index, so $Ax = x$ for every x. Hence $A = I$ and the final wire labels are again $e_1, \ldots, e_m$.

For any nonzero-valued function $h : \mathbb{F}_2^m \to \mathbb{C}$ and any label v, define

$$\mathcal{I}_v(h) = \prod_{x \in \mathbb{F}_2^m} h(x)^{(-1)^{v \cdot x}}.$$

The exponents are integers $+1$ or -1, so negative exponents mean multiplicative inverses, not a choice of logarithm or phase branch. In particular, $\mathcal{I}_v(h_1 h_2) = \mathcal{I}_v(h_1)\mathcal{I}_v(h_2)$.

Lemma 1. *If $v \neq 0$, then $\mathcal{I}_v(c) = 1$ for each constant $c \neq 0$. If also $w \neq 0$ and $w \neq v$, then $\mathcal{I}_v(x \mapsto g(w \cdot x)) = 1$ for every nonzero-valued $g : \mathbb{F}_2 \to \mathbb{C}$.*

Proof. Choose u with $v \cdot u = 1$. Pairing x with $x + u$ cancels the constant factors in the first assertion. For the second assertion, the distinct nonzero vectors v, w are linearly independent over $\mathbb{F}_2$. Therefore the map $x \mapsto (v \cdot x, w \cdot x)$ is onto $\mathbb{F}_2^2$, and there is a u with $v \cdot u = 1$ and $w \cdot u = 0$. The same fixed-point-free pairing preserves $g(w \cdot x)$ and reverses its exponent. Each pair contributes 1. □

Call v even when $\mathrm{wt}(v)$ is even, equivalently $v \cdot \mathbf{1} = 0$. The diagonal function $d_*(x) = e^{\mathrm{i}\theta_*(x)}$ is i at $0, \mathbf{1}$ and 1 elsewhere. For every nonzero even v,

$$\mathcal{I}_v(d_*) = \mathrm{i}\,\mathrm{i} = -1,$$

since both marked strings have dot product zero with v. If v never occurred on a rotated wire, Lemma 1 would give $\mathcal{I}_v(f) = 1$. But $f = cd_*$ for a realizing circuit, so it also gives $\mathcal{I}_v(f) = \mathcal{I}_v(c)\mathcal{I}_v(d_*) = -1$, a contradiction. Thus *every nonzero even label must occur on a rotated wire.* This argument is insensitive to phase wrapping.

3 Counting even outputs and returns

Let $H = \{v \in \mathbb{F}_2^m : v \cdot \mathbf{1} = 0\}$. It has dimension $m-1$, and contains $2^{m-1}-1$ nonzero vectors. All initial wire labels are odd. The first appearance of any even label must therefore be a CNOT output; rotations do not change labels. Since a single CNOT produces one output label, at least $2^{m-1}-1$ CNOTs have even output.

An update whose target and control labels are both odd is a *birth*; its output is even. At every stage, every even label ever produced lies in the span of birth outputs produced so far. To prove this invariant, initially there are no even labels. An odd–odd update adds its birth output as a generator. An even–even update outputs the sum of two labels already in the span. The two mixed-parity updates produce odd labels, so they add no even label. Unchanged rows preserve the invariant. This proves the assertion inductively, including labels that later disappear from the current rows.

All nonzero vectors of H must be produced, so the birth outputs span H. If B is the number of births, this forces $B \geq m-1$. Now count the number E of even wire labels currently present:

Target before	Control	Output	Change in E
odd	odd	even	$+1$ (birth)
even	odd	odd	-1 (return)
even	even	even	0
odd	even	odd	0

The initial and final labels are standard basis rows, so E starts and ends at zero. Hence the number R of returns equals B, and $R \geq m-1$. Returns have odd output and are disjoint from the even-output CNOTs already counted. Therefore the total CNOT count is

$$K \geq (2^{m-1}-1) + R \geq (2^{m-1}-1) + (m-1) = N_m.$$

Together with the complement invariance of θ_*, this proves Theorem 1.

Corollary 1 (Using the literature upper bound)**.** *In the stated model, the worst-case optimal CNOT count among all reflection-symmetric diagonal unitaries on $2n$ wires is N_{2n}.*

Proof. Algorithm 1 of Huang et al. [1], Sect. 4 and Appendix C, implements every target in the class up to global phase with $(2^{2n-1}-1) + (2n-1) = N_{2n}$ CNOTs. This is the literature upper-bound premise. Theorem 1 supplies a target meeting it from below. □

Higher-dimensional count and remaining questions. For integers $d, n \geq 1$, substituting dn for n gives the exact worst-case count $2^{2dn-1} + 2dn - 2$ on $2dn$ wires. In particular, the $6n$-wire count $2^{6n-1} + 6n - 2$ in Sect. 6 of Huang et al. is sharp for the full class of complement-symmetric diagonal targets. Their three-dimensional interaction-potential family, defined by Eq. (5), has additional radial symmetry: the lower bound here does not establish worst-case optimality within that narrower family. Section 6 also asks for optimal circuit depth. The gate-count proof does not determine that depth, since disjoint gates can run in parallel. Thus the depth and restricted-potential synthesis questions remain unresolved by this argument.

4 A finite synthesis check

With $\chi_v(x) = (-1)^{v \cdot x}$,

$$\theta_*(x) = \frac{\pi}{2^m} \sum_{v \in H} \chi_v(x).$$

This follows because a character sum over H is $|H|$ on $H^\perp = \{0, \mathbf{1}\}$ and zero elsewhere: outside $H^\perp$, translation of the summation index by a vector pairing nontrivially with x cancels terms. Rotating once at each nonzero even label by $-\pi/2^{m-1}$ thus implements $e^{-\mathrm{i}\pi/2^m} D_*$ whenever the labels return to identity.

Exact breadth-first searches over ordered wire labels and visited nonzero even labels give shortest identity-return networks of lengths 2 and 10 for $m = 2$ and $m = 4$, respectively. Inserting these rotations at each label's first occurrence gives circuits for D_* of those lengths. Integer and rational-arithmetic checks also verify every hard-instance character and every basis phase in the displayed identity for $m = 2, 4, 6, 8$. These are finite checks, distinct from the all-n written proof in Sections 2–3. The pinned declaration `ReflectionDiagonalCNOTLowerBound.result` proves the lower bound; worst-case equality additionally uses the cited synthesis upper bound. The character and phase checks cover $m = 2, 4, 6, 8$, and the shortest-network checks cover $m = 2, 4$. The exact checker and witnesses accompany the source package.

References

[1] X. Huang, T. Kosugi, H. Nishi and Y.-i. Matsushita, *Optimized Synthesis of Circuits for Diagonal Unitary Matrices with Reflection Symmetry*, Journal of the Physical Society of Japan **93**(5), 054002 (2024). doi:10.7566/JPSJ.93.054002. arXiv:2310.06676v2, Eq. (1), Algorithm 1, Sect. 6 and Appendix C.

[2] M. Amy, P. Azimzadeh and M. Mosca, *On the CNOT-complexity of CNOT-PHASE circuits*, published as *On the controlled-NOT complexity of controlled-NOT–phase circuits*, Quantum Science and Technology **4**, 015002 (2019), doi:10.1088/2058-9565/aad8ca; arXiv:1712.01859v2 (2018), Sect. 2.

[3] Y. Cao, Y. Lu, J. Nie, X. Sun and G. Tian, *Toward Minimum Graphic Parity Networks*, arXiv:2509.10070v1 (2025), Theorem 1.1.

[4] Trureturing repository contributors, version-pinned sources: ReflectionDiagonalCNOTLowerBound.lean and ReflectionDiagonalParityCounting.lean. The lower bound is `result : claim` in the first module; the second supplies `count_bound`.